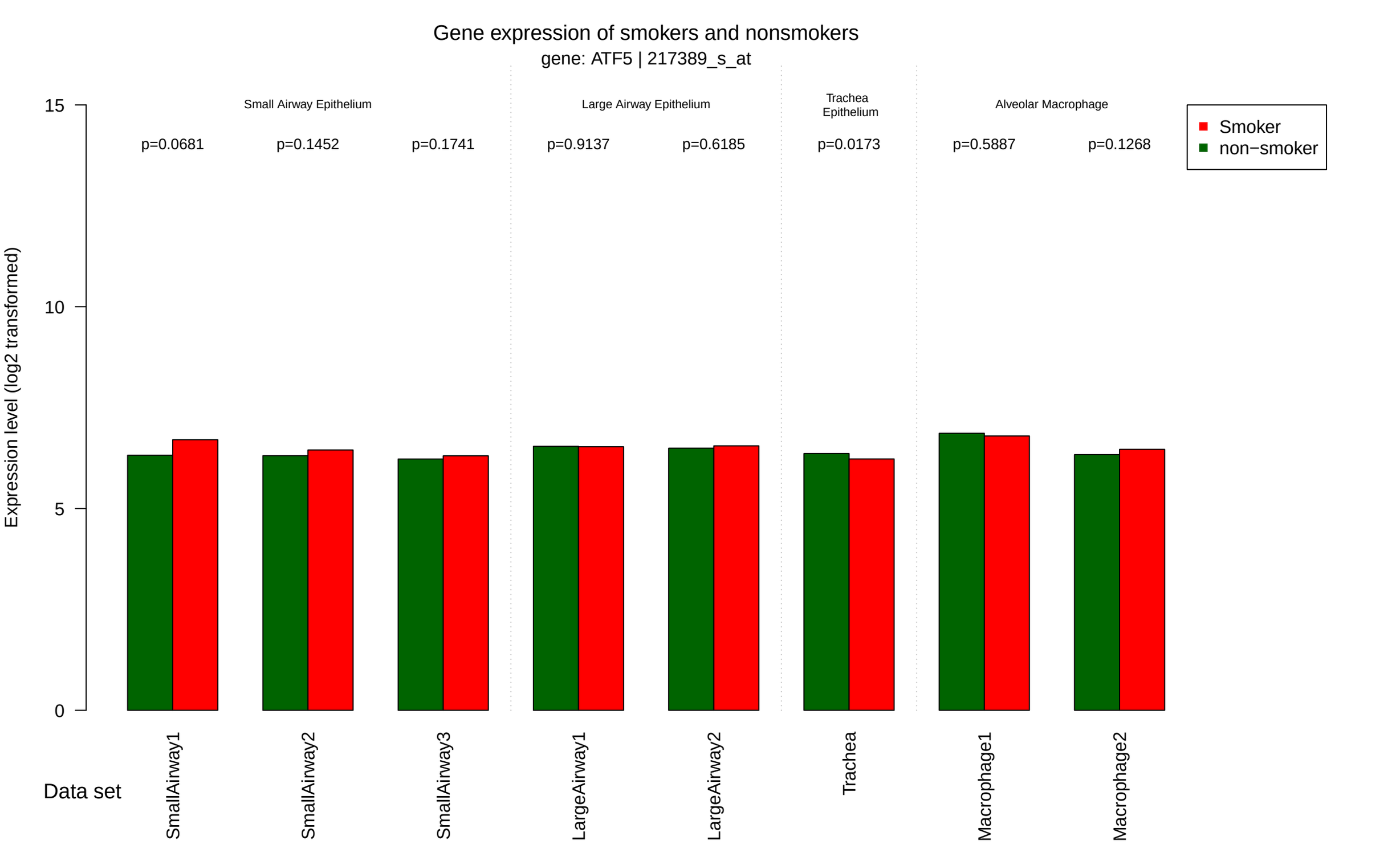
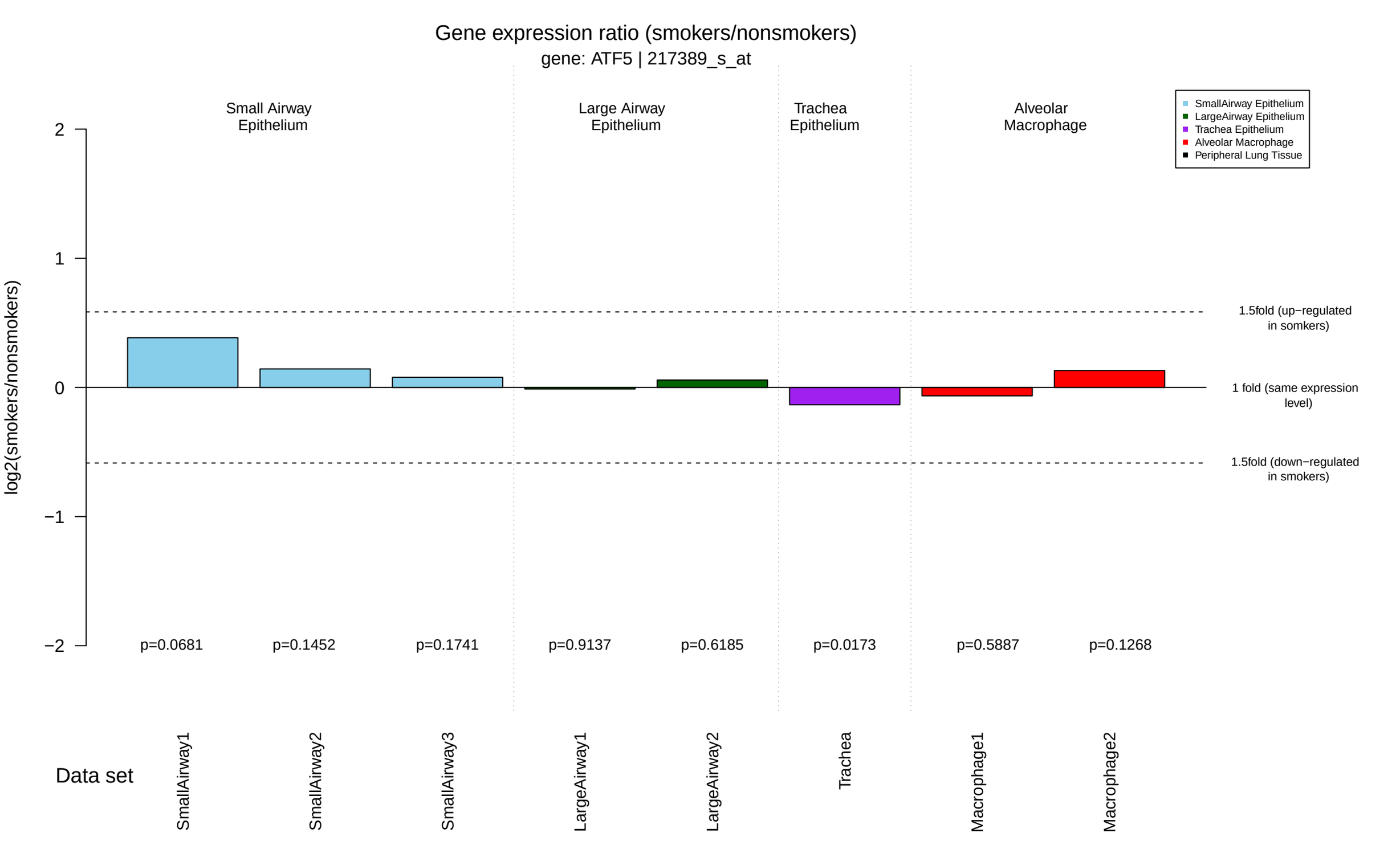
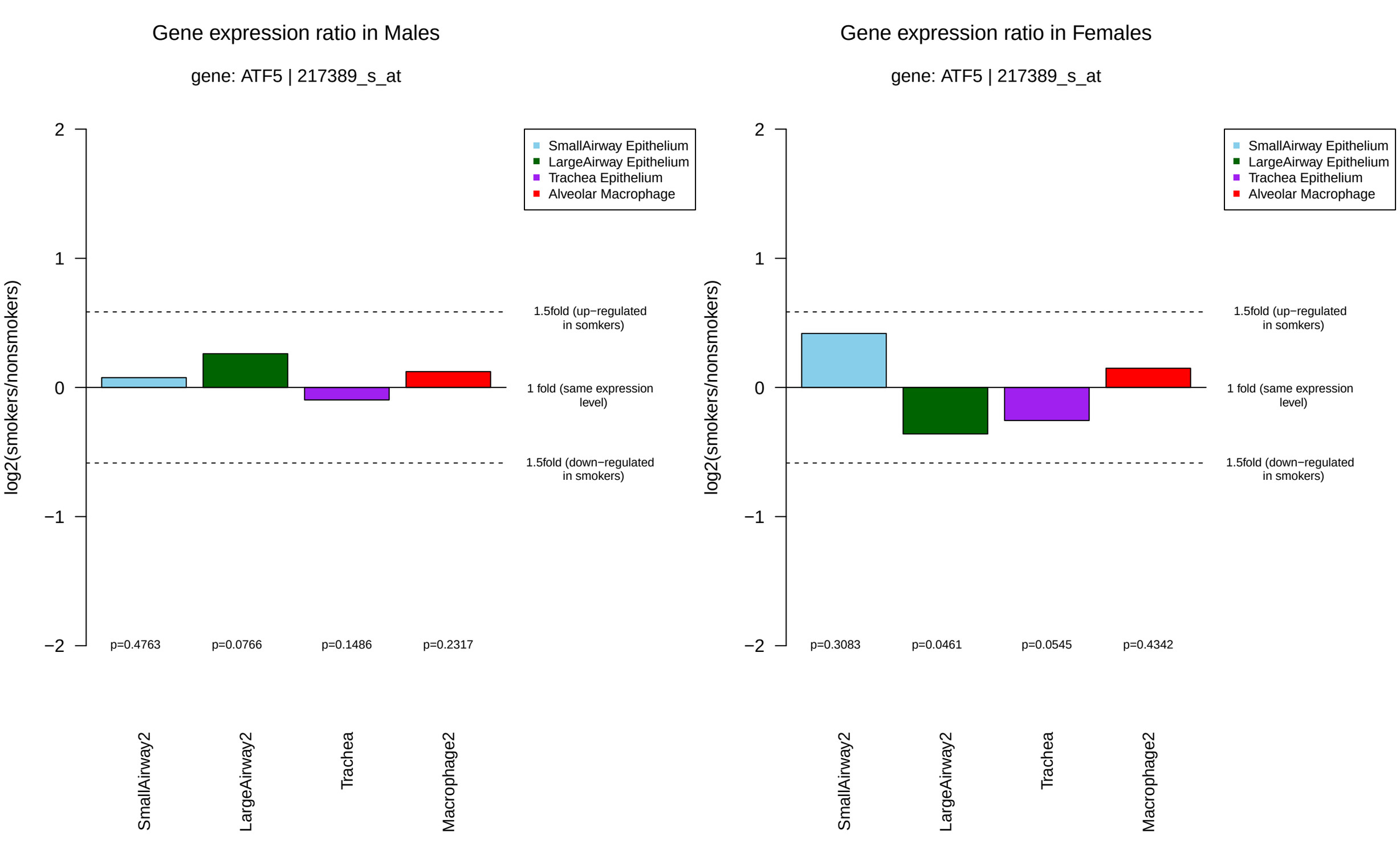
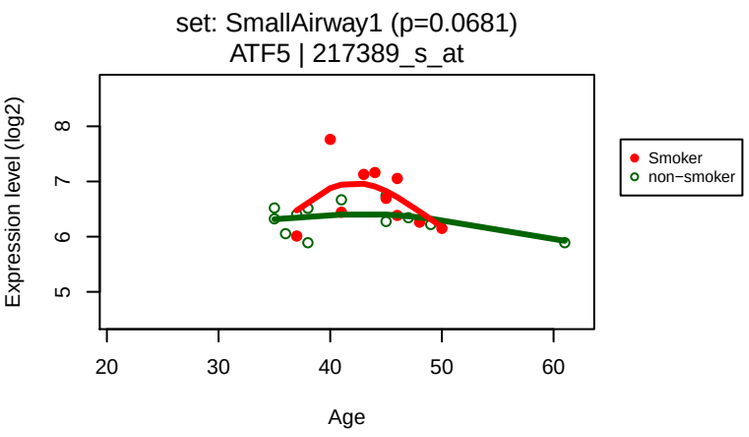
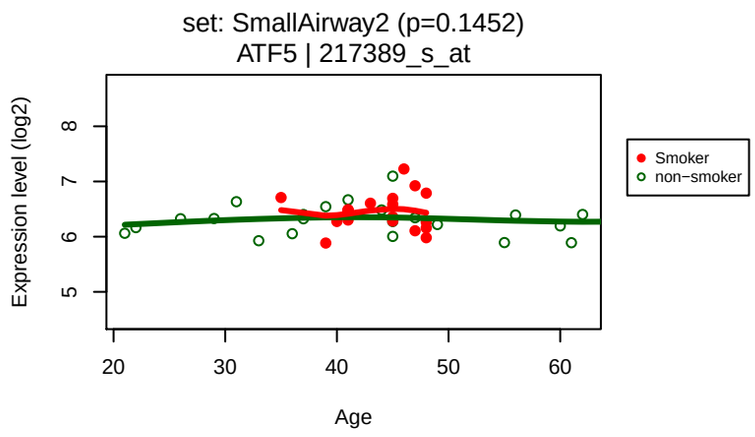
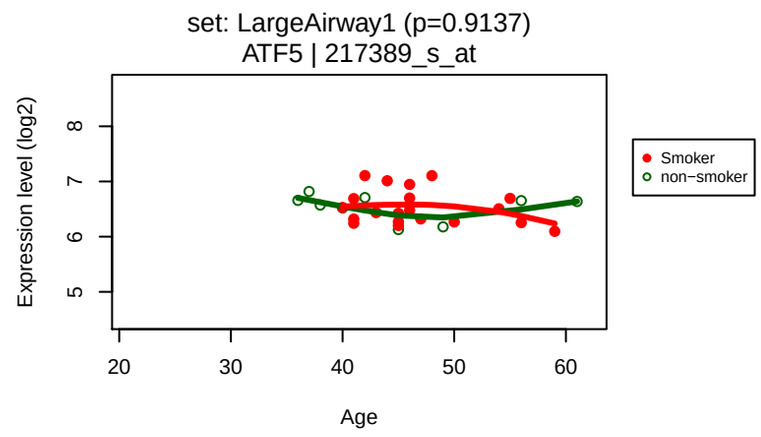

Gene expression of smokers and nonsmokers
gene: ATF5 | 217389_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0681
p=0.1452
p=0.1741
p=0.9137
p=0.6185
p=0.0173
p=0.5887
p=0.1268
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

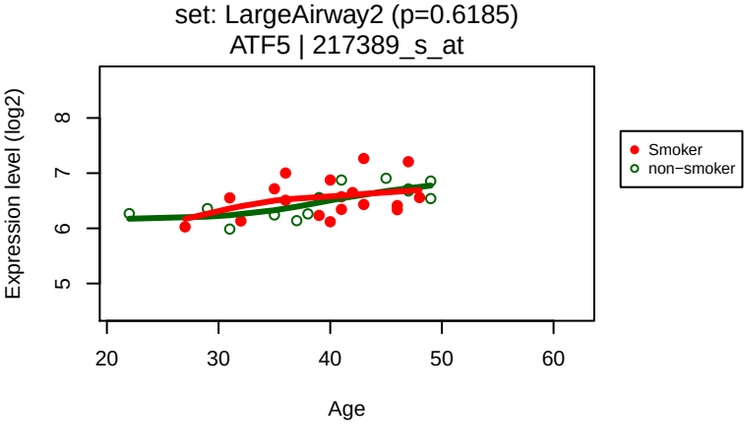

Gene expression ratio (smokers/nonsmokers)
gene: ATF5 | 217389_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0681
p=0.1452
p=0.1741
p=0.9137
p=0.6185
p=0.0173
p=0.5887
p=0.1268
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

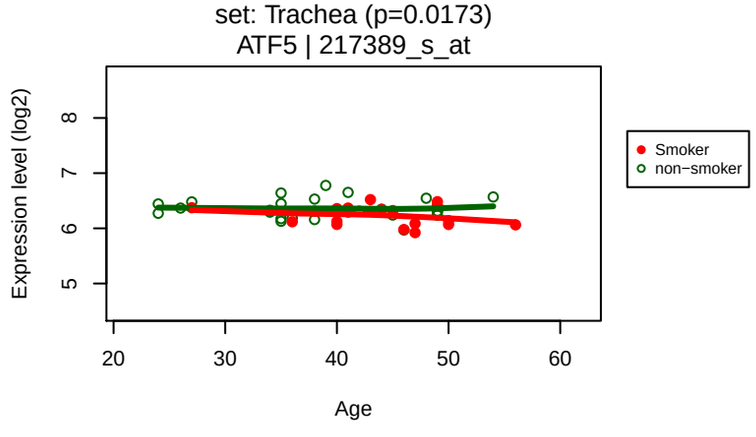

Gene expression ratio in Males
gene: ATF5 | 217389_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4763
p=0.0766
p=0.1486
p=0.2317
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATF5 | 217389_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3083
p=0.0461
p=0.0545
p=0.4342
SmallAirway2
LargeAirway2
Trachea
Macrophage2

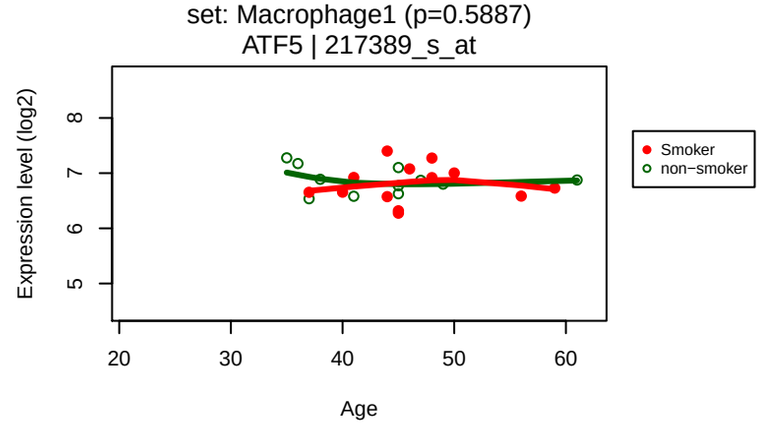

set: SmallAirway1 (p=0.0681)
ATF5 | 217389_s_at
Expression level (log2)
Age
Smoker
non−smoker

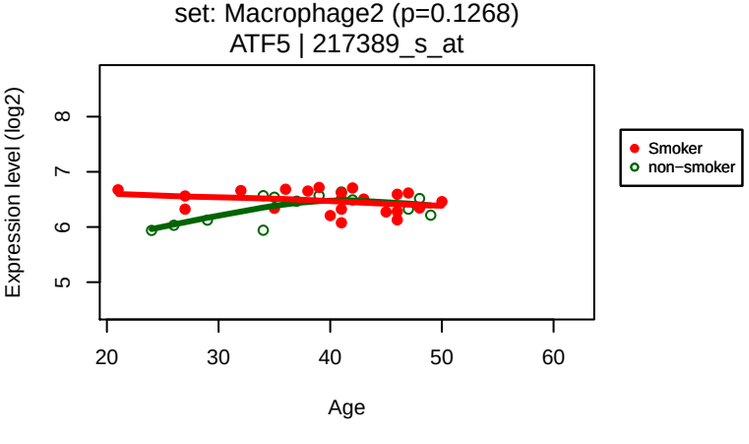

set: SmallAirway2 (p=0.1452)
ATF5 | 217389_s_at
Expression level (log2)
Age
Smoker
non−smoker

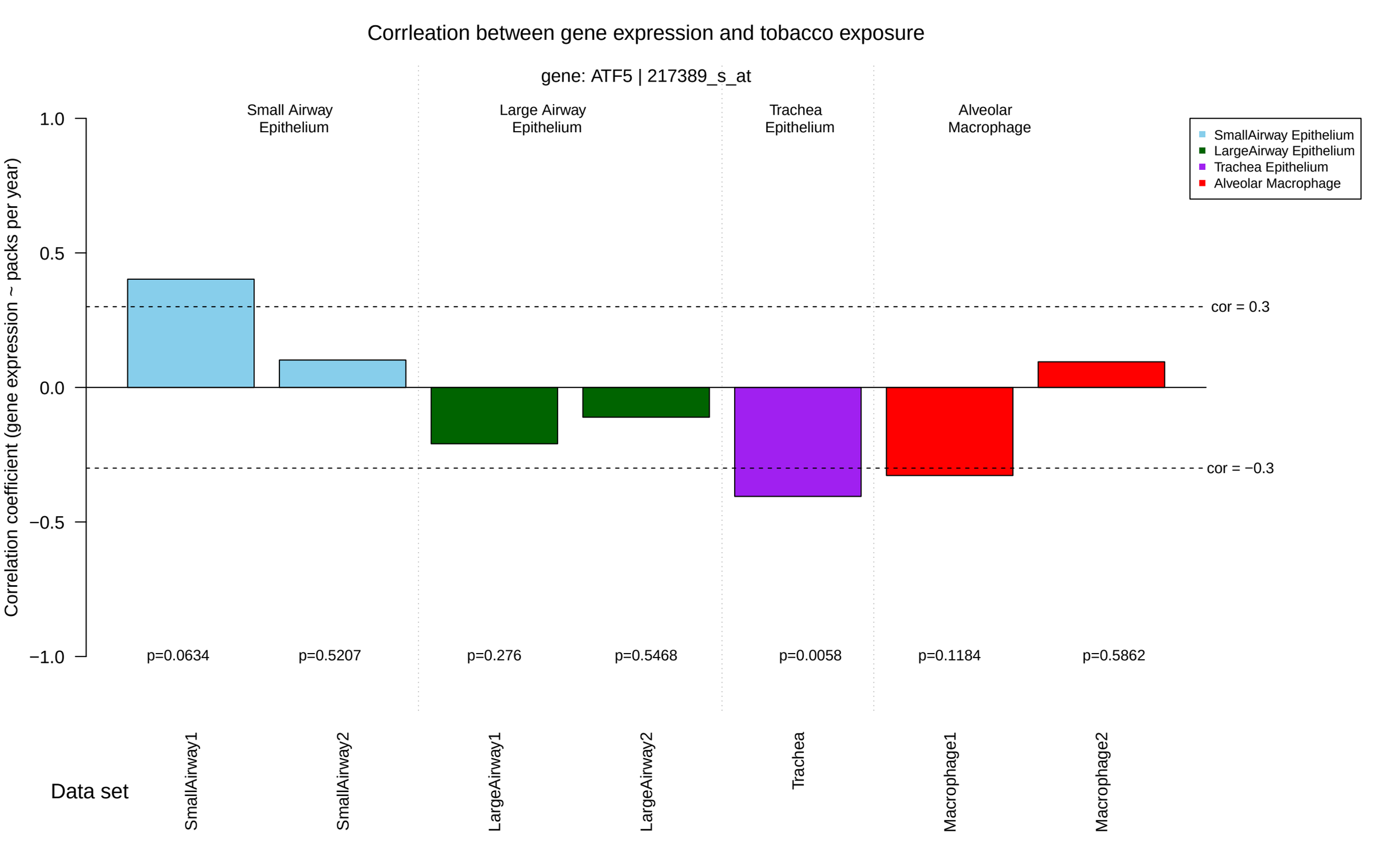

set: LargeAirway1 (p=0.9137)
ATF5 | 217389_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6185)
ATF5 | 217389_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0173)
ATF5 | 217389_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5887)
ATF5 | 217389_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1268)
ATF5 | 217389_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATF5 | 217389_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0634
p=0.5207
p=0.276
p=0.5468
p=0.0058
p=0.1184
p=0.5862
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2